

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 8

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 9 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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PHYS10302 Vibrations and Waves

Week 9 Questions

1. A piece of glass has a refractive index $n = 1.48$ for light of which the wavelength in vacuum is 500 nm and $n = 1.49$ for light of which the wavelength in vacuum is 600 nm. For each wavelength give
 - (a) the speed (phase velocity, v_p) in the glass;
 - (b) the wavelength in the glass;
 - (c) the wavenumber $k = 2\pi/\lambda$ in the glass;
 - (d) the angular frequency ω .
2. By drawing inspiration from the lecture on the method of solution for the 2D wave equation in Cartesian coordinates, suggest how we would solve the wave equation in general vector notation without specifying a specific coordinate basis (with $v = c$) by turning the PDE into an ODE for the time dependent part and a PDE for the spatially dependent part. Derive the PDE for the spatial part of the solution, i.e. Helmholtz's equation.
3. Explain what is meant by *normal* and *anomalous* dispersion. A ray of white light falls on a prism which produces a spectrum and the blue component is bent more than the red component. Is this normal or anomalous dispersion? Why?
4. Suppose the dispersion relation for waves on a string is given by

$$\omega^2 = \frac{Tk^2}{\mu} + \alpha k^4,$$

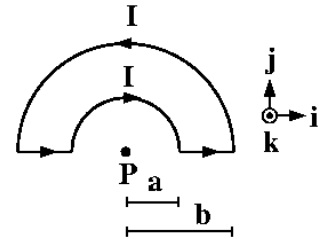
where T is the tension in the string, μ is the mass per unit length and α is a constant that depends on the string.

Find expressions for the phase and group velocities of waves on the string.

5. You may wish to recap the key points in the course so far, focussing on how we obtained the equations studied.

E&M Tutorial Sheet 8: Ampère's Law and more on Lorentz and Work

- 1) (Revision) Consider a wire as shown in the diagram consisting of a semi-circular loop of radius a , a concentric semi-circular loop of radius b , and straight wire sections joining the two semi-circles. The wire carries a current I . Using the Biot-Savart Law, calculate the magnetic field at the centre of the loops.

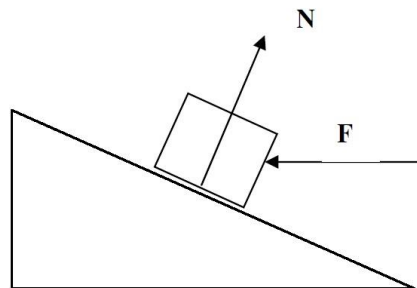


- 2) A steady current I flows down a long cylindrical wire of radius a . Find the magnetic field inside and outside the wire under the following conditions:
- When the current is uniformly distributed over the surface of the wire only.
 - When the current is distributed in such a way that the current density is proportional to the distance from the centre of the wire, r , i.e. $j(r) = kr$.

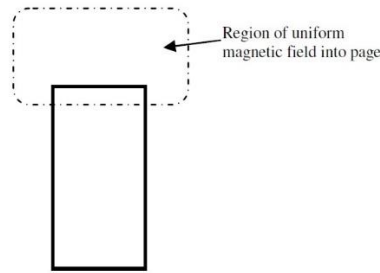
Hint: use Ampere's law with a pathway corresponding to a circular loop centred on the axis of the wire with radius $r < a$ for the inside, and $r > a$ for the outside. Carefully consider the amount of current passing through the area enclosed by the loop in each case.

The next question will look weird, but bear with me. It's for comparison with Q4, with the aim of drawing an analogy between an example in a familiar problem (in mechanics) and an example in magnetism. Both Q3 AND Q4 look long, but I only want you to provide QUALITATIVE answers. I'll illustrate my model answer with some formulae, but I want you to concentrate on thinking about physics involved.

- 3) A horizontal force, F , is being applied to an object to push it up a frictionless slope. The slope presents a normal reaction, N , and the arrangement is shown in the diagram below.



- Does the normal reaction do any work? Think about the direction the object moves in and the direction of the normal reaction.
 - What force is lifting the object? Think about which force has a component in the vertical direction.
 - What does the work?
 - Summarise in words the effect of the normal reaction in this problem.
- 4) A rectangular loop of wire with a total mass m hangs vertically with one end in a uniform magnetic field, which is into the page on the diagram below.



- a) What current must flow in the loop if the magnetic force balances gravity? Write down a formula just for this one part.

If more current flows, then the loop will rise upwards. A charge moving along the top of the loop then will have two components to its velocity. One, w , is just the horizontal motion associated with the flow of the current along the wire, $I = \lambda w$. The other is vertical, u , due to the loop rising upwards.

- b) Are magnetic forces providing the work to raise the loop? Be careful and don't just write down the first thing that comes to mind!
- c) Sketch the components of the velocity of a charge moving along the top of the wire. Add on your sketch the magnetic forces generated by each component of the velocity.
- d) Considering your answer to (c) what must be supplied to each charge in order to keep the current at a steady value? Suggest a component that could be added to the system to achieve this.
- e) What provides the work associated with your answer to (d)?
- f) Summarise in words the effect of the magnetic forces.
- g) Compare the situation to Q3.

PHYS10372 Mathematics 2: Tutorial-08 Questions

These questions cover the material from week-8 of PHYS10372: basis vectors, projections, transformation matrices, quantum states.

1.

- a. Find the projection of $\mathbf{a} = \begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$ that is parallel to $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. Determine also the perpendicular projection.
- b. Express the vector $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ in the basis given by $\mathbf{i}' = \frac{1}{\sqrt{2}}(\mathbf{i} - \mathbf{k})$, $\mathbf{j}' = \frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{k})$, $\mathbf{k}' = \mathbf{j}$.

2.

- a. A 2D vector-space is defined by two basis vectors: $\hat{\mathbf{e}}_1 = \frac{1}{5}(3\mathbf{i} + 4\mathbf{j})$ and $\hat{\mathbf{e}}_2 = \frac{1}{\sqrt{5}}(\mathbf{i} + 2\mathbf{j})$.
- Sketch these two vectors.
 - Show that the two vectors do not form an orthonormal basis.
 - Determine a basis vector that forms an orthonormal basis with $\hat{\mathbf{e}}_1$.
- b. A 3D vector-space is defined by the basis vectors: $\mathbf{e}_1 = (\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})$, $\mathbf{e}_2 = (\mathbf{i} - 4\mathbf{k})$ and $\mathbf{e}_3 = (5\mathbf{i} + 2\mathbf{j})$
- Show that these 3 vectors are independent but not orthogonal.
 - Use the Gram-Schmidt procedure to form an orthogonal basis $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ where $\mathbf{u}_1 = \mathbf{e}_1$.
 - Normalize the orthogonal basis to obtain an orthonormal basis.

3. The vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ is transformed by the operation of three matrices:

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}, \quad C = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

to the vector $\mathbf{v}'$ i.e. $\mathbf{v}' = CBA\mathbf{v}$. The transform is applied on the set of vectors that trace the circumference of a circle of unit radius.

- What transformation do each of these matrices perform.
- Sketch the shape defined by the transformed points after each transformation.
- By matrix multiplication, determine the matrices $M = DA$, where $D = CB$ and $N = EC$ where $E = AB$.
- Sketch the shape produced by the ABC transformation acting on the set of circle points.

4.

- A quantum system is in the state: $|\psi\rangle = 0.8|0\rangle - 0.6i|1\rangle$. What are the probabilities that the measurement finds the system in configuration $|0\rangle$ or in configuration state $|1\rangle$.
- Normalise the two quantum states: $|\psi_1\rangle = |0\rangle - |1\rangle + 2|2\rangle$, $|\psi_2\rangle = 2i|0\rangle - 2|1\rangle + |2\rangle$
- Write down a quantum description of a 6-sided dice with configuration states: $|1\rangle, |2\rangle, \dots, |6\rangle$. Write down a quantum description where the roll of the dice is twice as likely to give an odd number compared to an even number.
- Calculate the overlap $\langle\psi|\phi\rangle$ for $|\psi\rangle = \frac{2}{3}|0\rangle + \frac{\sqrt{5}i}{3}|1\rangle$ and $|\phi\rangle = \frac{1}{\sqrt{5}}(|0\rangle - |1\rangle + i\sqrt{3}|2\rangle)$.
- A quantum system is initially in the state: $\frac{1}{\sqrt{12}}(|0\rangle + |1\rangle - |2\rangle - 3|3\rangle)$. The system undergoes the transformation: $|0\rangle \rightarrow |3\rangle, |1\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), |2\rangle \rightarrow |2\rangle, |3\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.

- i. Write this transformation as a matrix equation of the form $\boldsymbol{v}' = \boldsymbol{M} \boldsymbol{v}$ where $\boldsymbol{v}$ and $\boldsymbol{v}'$ are the states before and after the transformation, respectively and $\boldsymbol{M}$ is the transformation matrix.
- ii. What is the state of the quantum system after this transformation.
- iii. Calculate the overlap (inner-product) of the initial state of the system with the state after the transformation.
- iv. What is the probability of measuring the system in state $|0\rangle$ after the transformation.